

Santosh Kumar Chaudhary

Electrical & Electronic Engineering

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering with hands-on exposure to power system operation, protection, automation, and fault analysis, plus practical experience in app development and Firebase-based monitoring systems. Interested in transmission, distribution, substations, automation, and engineering consulting roles where analytical skills, simulation work, and field learning can support reliable power infrastructure projects. Experienced in MATLAB/Simulink-based modeling, ANN-based fault detection, Android application development, and real-time monitoring system design.

CORE SKILLS

Power & Engineering Power system analysis, fault detection, electrical circuit design, transmission and distribution fundamentals	Software & Tools MATLAB/Simulink, AutoCAD, ETABS, Excel, Firebase Realtime Database, Work Manager
Programming Python, Java, C	Data & AI Artificial Neural Networks (ANN), data analysis, machine learning, technical problem solving

EDUCATION

- Nepal Engineering College (nec), Changunarayan** - Bachelor's in electrical and Electronic Engineering
- Kathmandu Model College, Bagbazar** -+2 Science
- Little Angels' School, Hattiban, Lalitpur** - School Level

EXPERIENCE

Intern - Nepal Electricity Authority (NEA) | 1 Month Industrial Training

- Gained practical exposure to power system operation, maintenance, and distribution activities.
- Observed protection coordination, load management, and system monitoring processes.
- Assisted engineers with daily operational tasks, report preparation, and field data collection.

SELECTED PROJECT

Wavelet and ANN Based Fault Detection and Communication-Controlled System | Final Year Project

- Built an intelligent protection framework for power grids using Discrete Wavelet Transform (DWT) and Artificial Neural Networks (ANN) for rapid fault detection, classification, and isolation.
- Modeled and simulated the system in MATLAB/Simulink, processed voltage and current sequence components, and used wavelet detail coefficients as ANN input features.
- Connected the simulation to an Android monitoring application through Firebase Realtime Database for real-time status visualization and communication-assisted breaker control.
- Project tools and technologies: MATLAB/Simulink, ANN, DWT, Android Studio, Java, Firebase Realtime Database.

PUBLICATIONS

- ❖ **Multi-Class Fault Type Classification Using Sequence Components and Wavelet Features with Comparative Study of ML Models and Neural Networks** | IEEE International Conference on ICT and Photonics (ICTP 2026), Kathmandu, Nepal.
 - **Research area:** Power-system fault classification using sequence components, wavelet features, ML models, and neural networks.
 - **DOI:** [10.1109/ictp67998.2026.11485076](https://doi.org/10.1109/ictp67998.2026.11485076)
- ❖ **An Efficient Deep Learning Based Mobile Application for Pigeon Pea Disease Classification** | IEEE International Conference on ICT and Photonics (ICTP 2026), Kathmandu, Nepal.
 - **Research Area:** Deep learning, computer vision, mobile application development, plant disease detection, image classification, and AI-based agricultural disease diagnosis.
 - **DOI:** [10.1109/ictp67998.2026.11485179](https://doi.org/10.1109/ictp67998.2026.11485179)
 - **Recognition:** Best Paper Award.

LEADERSHIP & ACTIVITIES

- ❖ **IEEE Nepal Engineering College Student Branch Chairperson**
 - Leading technical workshops, seminars, and competitions while promoting innovation and professional development among students.
- ❖ **IEEE ACEI Ambassador**
 - Represented ACEI at the college level and conducted workshops and events on engineering, ethics, and AI awareness.
- ❖ **Secretary, CRAFT *nec***
 - Assisted in organizing and managing technical events and student competitions.

ADDITIONAL PROJECT

Earnify (Founder & Developer)

- Developed and published an ad-based earning application on the Google Play Store that enables users to earn Nepali currency.
- App link: [Earnify on Google Play Store](#)
- Integrated Firebase Realtime Database and Firebase Cloud Messaging for coin storage and automated notifications.
- Implemented WorkManager to dynamically fetch and schedule notifications from the admin dashboard.

CERTIFICATIONS & ACHIEVEMENTS

- Received the Best Paper Award at the IEEE International Conference on ICT and Photonics (ICTP 2026) for the paper titled “An Efficient Deep Learning Based Mobile Application for Pigeon Pea Disease Classification.”
- Won the UI/UX Competition at Nepal Engineering College Exhibitions.
- Secured runner-up position in the Software Competition at HEX.
- Ranked in the Top 7 in the GCP-Powered AI Solution for Building on Aelf, organized by StackUp.
- Became a finalist in STEAM, organized by the Ministry of Education, Science, and Technology, Nepal
- Participated in the Hult Prize.